

MULTIVITAMIN USE AND RISK OF CANCER AND CARDIOVASCULAR DISEASE IN THE WOMEN'S HEALTH INITIATIVE COHORTS

Context Millions of postmenopausal women use multivitamins, often believing that supplements prevent chronic diseases such as cancer and cardiovascular disease. **Objective** To examine associations between multivitamin use and risk of cancer, cardiovascular disease and mortality in postmenopausal women. **Design, Setting and Participants** 161,808 participants from the Women's Health Initiative Clinical Trials (n=68,132 in three overlapping trials of hormone therapy, dietary modification and calcium-vitamin D) or Observational Study (n=93,676). Detailed data were collected on multivitamin use at baseline and follow-up time points. Study enrollment occurred between 1993–1998; women were followed for a median of 8.0 years in the clinical trials and 7.9 years in the observational study. Disease endpoints were collected through 2005. **Outcome Measures** Cancers of the breast (invasive), colon/rectum, endometrium, kidney, bladder, stomach, ovary and lung; cardiovascular disease (myocardial infarction, stroke, venous thrombosis); and total mortality. **Results** 41.5% of participants used multivitamins. After a median of 8.0 years of follow-up in the CT and 7.9 years in the OS, 9,619 cases of breast, colorectal, endometrium, kidney, bladder, stomach lung or ovary cancer; 8,751 CVD events and 9,865 deaths were reported. Multivariate-adjusted analyses revealed no association of multivitamins with risk of cancer (breast HR=0.98, 95%CI 0.91–1.05; colorectal HR = 0.99, 95% CI 0.88–1.11; endometrial HR = 1.05, 95%CI= 0.90–1.21; lung HR = 1.0, 95% CI=0.88–1.13; ovary HR = 1.07, 95%CI =0.88–1.29); CVD (MI HR= 0.96, 95%CI= 0.89–1.03; stroke HR = 0.99, 95%CI =0.91–1.07; VT HR = 1.05, 95%CI =0.85–1.29); or mortality (HR = 1.02, 95% CI=0.97–1.07). **Conclusion** After a median follow-up of 8.0 and 7.9 years in the CT and OS, respectively, the WHI cohorts provide convincing evidence that multivitamin use has little or no influence on the risk of common cancers, cardiovascular disease or total mortality in postmenopausal women. Clinical Trial Registration clinicaltrials.gov identifier: NCT00000611